

FARMARE — Due Diligence Document

Automated DeFi yield strategy — Prudent 80/20 profile

Version 1.0 — 2026-07-09. Status: live paper trading (Phase 1). No client funds.

1. Executive summary

FARMARE is an automated yield system on Ethereum L2s (Base, Arbitrum) that combines two return streams under a strict risk framework:

- **CORE (80%)** — stablecoin yield across whitelisted blue-chip protocols, diversified over two venues (max ~50% per protocol), with automatic rotation and **fixed-rate locking** (Pendle PT) when fixed beats floating.
- **ENGINE (20%)** — concentrated ETH/USDC liquidity provision (Uniswap v3) with a volatility-adaptive range and a **conditional short-perp hedge** that converts the position from a directional bet into a fee-carry trade; it switches to pure delta-neutral funding capture when funding clearly pays more than LP fees.

Validated expectations (net of all modeled costs): portfolio +7.5% APY in the worst tested regime (a real semester in which ETH fell 44%), +10–16% in flat/bull regimes; max drawdown $\leq$ ~1% at portfolio level in the historical window. Engine-only long-history validation (2023–2026): +13% to +22% each calendar year. **These are simulated results** — see §7.

2. Where the yield comes from (and why it persists)

1. **Stable lending / LP spreads** — money-market and AMM equilibrium rates (5–12% on-chain). Persistent because on-chain credit demand pays it.
2. **Fixed-vs-floating premium** — PT buyers are paid 2–6pp to absorb rate compression risk. Persistent because most capital chases headline APY.
3. **LP fees minus hedged risk** — concentrated liquidity earns outsized fees; the vol-adaptive range plus delta hedge removes most of the directional and gamma cost. Persistent because passive LPs systematically under-manage this trade-off.
4. **Perp funding** — longs pay shorts most of the time; captured only when it clearly beats the alternatives.

3. Outcome range by regime (backtested, net of costs)

The strategy is *selected* on its worst regime — but most regimes are not the worst. The full validated range:

Market regime	Portfolio (Prudent 80/20)	Engine sleeve
Bear — real semester, ETH –44%	+7.5% APY (the floor)	+17.6% APY
Sideways (synthetic, mean of paths)	+16.3% APY	+28.5% APY
Bull (synthetic, +60% drift)	+10.0% APY	+8.1% APY
Engine, full period 2023–2026	—	+22.5% APY , every year positive
Decade 2016–2026, portfolio view (guarded engine)	—	positive every year , worst +3.7%

Upgrades adopted after these tests (PT fixed-rate locking: +5.9pp on the core evidence window; funding-carry mode: +3.4pp on the bull-regime engine) are additive to these figures and are being confirmed by the live forward test before being claimed.

4. Evidence base (all reproducible from this repository)

Test	Data	Result
Engine, real bear semester (ETH –44%)	Real pool volumes, prices, funding	Hedged LP +26% vs hold , –5% maxDD; naked LP –23% vs hold (rejected)
Three market regimes (bear real, flat, bull)	Real + synthetic	Prudent 80/20 dominates: never worse than –2pp vs best alternative in ANY regime
Three chains (Base, Solana, BSC)	Real data, same strategy	Hedge thesis confirmed on 3 chains / 3 DEXes / 3 assets
Long history 2023–2026	Binance prices, HL funding, calibrated volumes	Engine positive every calendar year (+13% to +22%), incl. +90% bull and –42% bear years
Fixed-rate (PT) vs floating core	147d of Pendle implied rates	PT ladder +11.5% vs +5.6% floating
Concentration cap	336d stable-pool APY histories	Halving protocol concentration costs 1.95pp APY — accepted for client readiness
On-chain fee-model calibration	Base RPC, live	Measured competing liquidity 0.49–0.53 vs 0.60 assumed → model is conservative
Stress suite (7 scenarios)	Injected into production code	7/7 pass — see §5
Decade test 2016–2026	Coinbase prices, era-anchored fee economics	Unguarded engine dies in 100%+-vol eras → viability guard adopted ; guarded portfolio positive every year (worst +3.7%)
Rejected by testing	—	naked tight ranges, vol-based regime switching, SMA trend filters, venue rotation, expected-yield sizing, stable looping (each documented in backtester/README.md)

Methodology discipline: every strategy claim was adversarially backtested with a **maximin criterion** (optimize the WORST regime, not the average); four intuitive ideas were rejected by the data. Two model bugs were found and fixed during development (fee-share width scaling; both documented).

5. Risk framework

Whitelist. Funds touch only: Aave, Morpho, Fluid, Pendle, Compound, Curve, Aerodrome, Uniswap. No new protocol without explicit owner action.

Automatic guards (checked hourly, acted on within 6h):

- Depeg: any underlying < \$0.99 → pool excluded; if held → emergency exit.
- TVL collapse: –30% vs 7-day high → excluded / emergency exit.
- Diversification invariant: the two core slots never share a protocol.
- Engine data sanity: absurd price/TVL aborts the run (no decisions on bad data).
- Fee-model drift: on-chain measurement vs assumption, alert at ±25%.

- Margin stress: fast ETH rally with a short open → alert (policy: margin buffer = 40% of short notional; forced de-risk at 50% consumption).
- **Engine viability guard** (decade-tested): the sleeve PARKS at stable yield when expected LP yield (fees – volatility drag) falls below 8% (re-entry above 15%), and kill-switches instantly on 14d vol > 100% or 30d momentum > +40% (parabolic markets). The engine only plays when the economics work — active ~29% of the 2016–2026 decade, ~60% of 2023–2026.

Quantified worst cases (stress suite, production code):

- Single-candle –35% ETH gap: $\approx -12\%$ on the engine sleeve = $\approx -2.4\%$ **portfolio** (residual short-gamma; cannot be hedged linearly).
- Underlying depeg during a PT lock: emergency exit fires, lock broken, cost $\approx 23\text{bps}$ of the slot + realized depeg drift until the next run ($\leq 1\text{h}$ detection, $\leq 6\text{h}$ action).
- Total API failure: run aborts with alert; state untouched; next run catches up hour-by-hour (no gaps in accounting).

Failure-mode table

Risk	Mitigation	Residual
Protocol exploit	Whitelist + 2-slot cap	$\leq \sim 40\%$ of portfolio per protocol (target: lower via more slots at scale)
Stable depeg	Hourly guard + auto-exit	$\sim 1\text{--}6\text{h}$ of exposure to the drift
ETH gap through range	Hedge + adaptive range	$\approx 2.4\%$ portfolio per -35% gap
Hedge venue failure	Margin policy, alerts	Unhedged LP until manual action
Model error	On-chain daily calibration check	Alert at $\pm 25\%$ drift
Operational (scheduler dies)	Cloud CI + watchdog + git audit trail	Missed accrual only; positions are static between runs

6. Operations & auditability

- Runs 4×/day in GitHub Actions (cloud, machine-independent); hourly guard checks; every decision appended to `paper/journal.jsonl`; **state committed to git after every run — the track record is immutable and timestamped.**
- Wealth-manager-grade Telegram reporting: every movement of capital reported with amount, source, destination, reason and cost.
- Dashboard (`docs/dashboard.html`) regenerated from state on every run.
- Secrets in CI vault; no private keys anywhere (Phase 1 is virtual).

7. Limitations — read before believing the numbers

1. **Simulated results.** Phase 1 is paper trading on live data: fills, slippage and gas are modeled (conservatively), not incurred.
2. **Model assumptions.** Fee share uses a calibrated competing-liquidity fraction (measured 0.49–0.53 vs 0.60 assumed — conservative direction); long-history volumes are a calibrated proxy ($\pm 2\text{pp}$ APY error bar measured on the overlap window); historical TVL held at today's level.
3. **Short histories.** PT-vs-floating evidence covers 147 days; the live forward test is days old. The acceptance gates below exist because of this.

4. **Capacity.** Strategy capacity is bounded by pool depth (engine fee share falls as position size approaches competing liquidity). Current design comfortable to ~\$1–2M; beyond that, multi-pool/multi-chain engine deployment (validated on Solana) is required.
5. **Perp margin** is modeled as a policy, not yet exercised with a real exchange account.
6. **The decade test is a counterfactual.** CLMMs did not exist before May 2021: the 2016–2022 legs run today's fee economics over historical price paths. Per-year digits carry wide error bars; the robust finding is structural (when the engine is unviable it must park — and the guard that enforces this is now in production).

8. Fund mechanics (built and tested, dormant until authorized)

Share-based accounting identical to the target ERC-4626 vault: NAV per share, high-water mark, entry/management/performance fees minted to the manager account. Deposits/withdrawals at daily NAV. Tested end-to-end in paper (deposit → fee accrual → withdrawal).

9. Regulatory status — the honest paragraph

Accepting third-party funds with discretionary management and fees is a **regulated activity** (MiCA in the EU). This system does not custody or accept client money today, and must not until: (a) legal structuring advice is obtained, and (b) the Phase 2 vault (self-custodial, non-discretionary smart contract) or an authorized wrapper is in place. Everything in this document is preparation material for that conversation — not an offer.

10. Acceptance gates before real capital

Gate	Threshold	Status
Forward test duration	≥ 4 weeks	running (started 2026-07-08)
Forward vs backtest	realized APY within model error bars for the regime	pending
Fee-model drift	7d avg within ±25% of assumption	passing (–12%)
Stress suite	7/7 green on every release	passing
Phase 2 vault	ERC-4626 + fork tests + external review	not started
Legal review	jurisdiction structuring for third-party funds	not started

Bottom line for an investor or bank: the strategy logic is validated across regimes, chains and 3.5 years of data; the risk framework is implemented and stress-tested in the production code; the track record is being generated in an auditable, tamper-evident way. What stands between this and real client money is time (forward-test maturity), the on-chain vault, and legal structuring — all scheduled, none improvised.